Detection of Personality Disorders based on Narratives

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***Abstract*—While computational methods have been widely applied to detect linguistic markers of mental health conditions such as depression and anxiety, Attention Deficit Hyperactivity Disorder (ADHD) has received comparatively little attention. This TFM addresses this gap by introducing a novel multimodal corpus of social media texts from individuals with ADHD and neurotypical controls. Using this resource, we conduct a comparative linguistic analysis and establish a set of baselines across three classification paradigms: Bag-of-Words, fine-tuned BERT, and zero-shot multimodal LLaVA-NeXT. Performance improves as we increase the model's complexity, yet all variants achieved limited overall performance. Different analyses show that users with ADHD posted more frequently, showed higher negative affect, and expressed less positive sentiment.**

***Index Terms*—Attention Deficit Hyperactivity Disorder, social media, Reddit, natural language processing, language models, text classification, multimodal analysis, BERT, corpus construction, linguistic markers**

I. Introduction

LANGUAGE offers a lens into the human mind. How people express themselves, the vocabulary and structures they employ, and the tone of their narratives reveal aspects of their personality and emotional state [1]. In recent years, advances in natural language processing and large language models have opened up new possibilities for analyzing these patterns for clinical purposes [2].

The motivation for leveraging computational methods in mental health research arises from a fundamental shift in how human experience is recorded. As Zhai and Massung [3] note, text serves as "the most natural way of encoding human knowledge," making it the primary medium for daily communication, from the trillions of emails sent to the countless blog articles, forum posts, and social media updates generated constantly. This explosion of online text has turned social media platforms like Reddit, Twitter (now X), and Facebook into a vast corpus of human thought and behavior. Because text is also "the most expressive form of information," capable of conveying complex psychological states, opinions, and personal narratives, these platforms offer an unprecedented lens through which to observe mental health phenomena at a population scale.

Traditionally, computational mental health research has focused on identifying linguistic patterns associated with depression and anxiety, yielding promising results in early detection and risk assessment [4]. However, Attention Deficit Hyperactivity Disorder (ADHD), one of the most prevalent neurodevelopmental disorder, affecting approximately 13.9% of adults [5] and 12.9% of children [6] in the United States, has received comparatively little attention from the NLP community. ADHD is characterized by persistent patterns of inattention, impulsivity, and hyperactivity that significantly impair daily functioning across educational, occupational, and social domains [7]. Many of the defining behavioral symptoms of ADHD manifest in language: excessive talkativeness, difficulty with turn-taking in conversation, tangential speech, and problems with narrative organization [8]. This makes language-based analysis a particular promising, yet under-explored, avenue for studying the disorder.

Social media platforms offer a unique opportunity for ADHD research for several reasons. First, individuals with ADHD often turn to online communities for peer support or information sharing [9], generating a rich corpus of natural language. Second, the informal and self-directed nature of social media posting may amplify the very linguistic features that clinical research has identified as characteristic of ADHD. Third, computational analysis of such data enables the study of ADHD at a scale that traditional clinical methods cannot easily achieve.

To address this gap, the present study is guided by the following research questions: (1) *What linguistic features distinguish the social media language of individuals with ADHD from that of neurotypical controls?* and (2) *How do different computational models compare in their ability to classify ADHD users?*

This TFM makes the following contributions:

1) The construction of a labeled corpus of ADHD and control social media texts and images.
2) An analysis of linguistic features identified in the corpus.
3) A comparative evaluation of three classification paradigms: A traditional Bag-of-Words (BoW) classifier serving as a baseline, a fine-tuned BERT model and a multimodal zero-shot classifier.

II. Related Work

This section reviews prior research in two areas relevant to the present study: computational approaches to mental health from social media and linguistic patterns associated with ADHD.

A. Computational Mental Health and Social Media

Driven by the potential of the large quantity of natural language available, the intersection of social media analysis and mental health has emerged as a vibrant field of study. Early foundational work demonstrated that language use on social media could serve as a powerful proxy for an individual's psychological state. A landmark study in this vein is the work of De Choudhury et al. [10], who proposed a computational framework for measuring the "emotional landscape" of Twitter users. By grounding their analysis in the psychological "circumplex model" of affect—which characterizes moods along dimensions of valence and activation—they were able to identify over 200 distinct moods and relate them to social behaviors like network activity and conversational engagement. This work established the validity of using psychologically-grounded models on large-scale social media text to understand human affective states. Building on this foundation, De Choudhury et al. [4] shifted from general mood expression to the task of predicting the onset of Major Depressive Disorder (MDD). In this study, they used crowdsourcing to identify Twitter users who reported a clinical depression diagnosis via a standardized psychometric instrument. By analyzing behavioral attributes over the year preceding diagnosis, they built a statistical classifier that could estimate depression risk before reported onset.

Expanding beyond single-condition analysis, Coppersmith et al. [11] demonstrated the potential for large-scale, multi-condition mental health research using social media language. Motivated by the "lack of data available to guide research" in mental health, they created a large dataset by identifying self-reported statements of diagnosis across ten distinct mental health conditions—including ADHD—on Twitter. This work establishes the methodological validity of using self-reported diagnoses as an alternative to clinically obtained data.

A parallel line of work has been advanced through the eRisk shared task series, which has provided a standardized benchmark for early risk detection of mental health conditions from social media since 2017, addressing tasks including depression and anorexia detection, or self-harm risk estimation, using Reddit data and emphasizing early detection. A survey of the first five years [12] concludes that while the tasks are feasible, performance remains modest (F1 scores typically below 0.70) and no single approach consistently dominates. The authors stress that these systems are not yet reliable enough for clinical use but show promise as scalable screening tools.

B. Language and Attention Deficit Hyperactivity Disorder

The relationship between language and ADHD has been investigated from clinical, neuropsychological, and, more recently, computational perspectives. In a review of the clinical literature, Tannock [8] synthesized findings from psychiatry, cognitive psychology, linguistics, and neuroscience to characterize the atypical language profiles associated with ADHD. A central insight from this review is that many of the defining behavioral symptoms of ADHD—including "excessive talkativeness," "blurts out answers," and "difficulty awaiting turn"—are themselves observable communicative acts, leading to the argument that ADHD may be partially defined by its linguistic manifestations.

C. Position of the Present Work

The present study builds on these foundations while addressing several gaps. Unlike prior work that classified ADHD as one condition among many using Twitter [11], we focus exclusively on ADHD with a richer per-user sample from Reddit, enabling finer-grained linguistic analysis. In contrast to the eRisk emphasis on early detection from minimal data [12], we leverage users' full posting histories. We also extend beyond text-only approaches by introducing a multimodal corpus and benchmark. Finally, while Tannock [8] establishes the clinical basis for ADHD-related language differences, our work provides a computational approach of these insights in a social media settings.

III. Methodology

A. Corpus Construction

For the construction of the corpus, we created a custom crawler to extract user-generated content from Reddit.

Reddit is a social media platform that thematically organizes its content into forums, or "subreddits." This structure, combined with its large user base and the publicly accessible nature of its content, makes Reddit a popular data source in prior computational mental health studies [12].

To enable a robust analysis of language use, the resulting dataset was designed to meet a set of criteria. The primary goal was to build a multimodal collection of data from two distinct user groups: individuals who self-identify as having ADHD and a control group without such a disclosure. The corpus was required to capture a complete view of user activity, including both textual data (posts and comments) and any associated images. Furthermore, the design prioritized the collection of rich metadata (e.g., timestamps, source) and the ethical protection of user privacy through an anonymization protocol.

1) Subreddit selection: For the selection of the subreddits to track, the need to obtain the following types of users was taken into account:

- **Users with ADHD**. Subreddits focused on ADHD.
- **Control users**. Subreddits not specific to ADHD, both mental health related and not.

It should be noted that the source subreddit does not serve as the criterion for labeling a user as having ADHD or not, as both ADHD and control users may post in either category of subreddit. The positive labeling criteria will be explained later in Section III-A2.

In addition, although the ratio of ADHD in adult men and women is close to 1:1 [13], Reddit has a more male demographic [14]. Not only that, but men are overall more likely to generate content online [15]. Because of this, and because ADHD symptoms tends to present differently depending on the person's gender [13], *r/adhdwomen* was also included to reduce gender bias. Therefore, the chosen subreddits are the following: *r/ADHD*, *r/adhdwomen*, *r/adhdmeme*, *r/ADHDthriving*, *r/mentalhealth*, *r/self* and *r/CasualConversation*. We also record each user's source subreddit, allowing us to filter by origin (e.g., exclude users from *r/CasualConversation*, whose contributions may be too off-topic).

2) Labeling users with ADHD: To determine whether a user suffers from ADHD, the code reviews all collected content and checks whether any of the phrases from a predefined list are detected (e.g., "My ? ADHD diagnosis"), indicating that the user has referred to having or being diagnosed with ADHD. This same approach has been applied in prior studies—for instance, to identify users with depression [16].

This list was created with some seed phrases and it was expanded using generative AI (Google Gemini) to obtain variants. Question marks in regular expressions ("?") act as wildcards to capture variable phrasing, and the string "ADHD" is configured to match a range of common variants, including "ADHDau," "ADD/ADHD," and "attention deficit hyperactivity disorder." The complete phrase list is provided in the project repository (see Appendix A).

Each comment and post receives a tag indicating whether a phrase was detected. While these tags are preserved, allowing ADHD status to be recomputed with different phrase lists in the future, the flagged content is removed during training to avoid trivializing the classification task. Flagged content is retained for corpus analysis.

3) Data extraction: Algorithm 1 shows the crawler implementation for data extraction. The process begins by iterating through the subreddit list and fetching their posts. For each retrieved post, its validity is checked (e.g., check if the post comes from a moderator and if the author did not delete their account). For each retrieved user, their content history (posts and comments) is obtained. They are labeled as positive, both at user and content level, if the ADHD regex pattern is detected. Images attached to the content are also downloaded if available.

Algorithm 1 Reddit Crawler for ADHD Corpus Construction.

Input: List of subreddits $\mathcal{S}$, ADHD phrase patterns $\mathcal{P}$
Output: User dataset $\mathcal{U}$, Content dataset $\mathcal{C}$

$\mathcal{U} \leftarrow \emptyset$; $\mathcal{C} \leftarrow \emptyset$
Compile regular expression ρ from $\mathcal{P}$

for each subreddit $s \in \mathcal{S}$ **do**
  for each content c retrieved from s **do**
    $u \leftarrow$ author of c
    if u is valid **then**
      $hasADHD \leftarrow$ False
      for each item i in user history of u **do**
        if i is not duplicate and has valid content **then**
          $c \leftarrow$ extract metadata and text from i
          download media from c if present
          c.pattern $\leftarrow \rho$ matches c.text
          $hasADHD \leftarrow hasADHD \vee c$.pattern
          $\mathcal{C} \leftarrow \mathcal{C} \cup \{c\}$
        end if
      end for
      if $u \notin \mathcal{U}$ **then**
        $\mathcal{U} \leftarrow \mathcal{U} \cup \{(u, hasADHD)\}$
      else if $\neg\mathcal{U}[u]$.ADHD $\wedge\ hasADHD$ **then**
        $\mathcal{U}[u]$.ADHD $\leftarrow$ True
      end if
    end if
  end for
end for

return $(\mathcal{U}, \mathcal{C})$

4) Anonymization and ethical considerations: To protect user privacy, the script implements an anonymization process. Neither usernames nor the original content ID are stored in the output files. Instead, a 16-character code is generated for each user and each piece of content, using a Hashed Message Authentication Code (HMAC) function with the SHA-256 algorithm. The secret key required for this process is obtained from an environment variable, ensuring that reverse mapping is not possible without access to this key. This technique allows linking contributions from the same author without revealing their identity.

All downloaded images were used in their original form during training. For the public release of the corpus, however, two post-processing steps were applied. The first automatically detects and blurs faces to protect user anonymity[1]. The second identifies and flags sexually explicit content[2]. The published dataset therefore contains face-anonymized images and excludes any image flagged by the detection model.

Furthermore, the data collected is that which is publicly available on Reddit, and no interaction with users takes place. Access to adult content is permitted at the request level via an age verification cookie, and the code adheres to the platform's rate limits to minimize its impact.

B. Data Preprocessing and Cleaning

This section describes the procedures applied to the raw data to address missing records, standardize text format, and derive additional aggregated features. These steps were essential for removing noise and preparing the data for analysis.

1) Handling missing data: All content for which no user exists, for any reason (e.g., connection failures during scraping, crawler stopping before saving the user), is deleted. Images associated with this content are also removed, as well as any empty image folders.

After cleaning the data, any remaining empty text and users who may have lost all their content and images are deleted.

[1]The model `arnabdhar/YOLOv8-Face-Detection` was used [17].
[2]The model `Falconsai/nsfw_image_detection` was used [18].

2) Data cleaning: The data cleaning process begins by removing non-English content.

Next, within the text, the following elements are removed: the string "<image>"; links; mentions of users and subreddits (strings beginning with "u/" and "r/," respectively); spoiler tags (">!" and "!<"); and mentions of age and gender in the typical Reddit format (e.g., "(22M)").

3) Aggregated data: Following data cleaning, aggregated features are added to both datasets, alongside the originally collected text, media, and metadata (e.g., timestamp, source). The content dataset is enriched with character and word counts, while the user dataset receives the total, mean, median, and standard deviation of content length, differentiated by posts and comments.

C. Strategies for Classification

To evaluate the discriminative power of linguistic features for ADHD detection, we adopt and compare three classification strategies. Each operates on the preprocessed textual data described in Section III-B and addresses the binary classification task of predicting whether a user belongs to the ADHD or control group.

1) Bag of Words: As a baseline, we employ a Bag-of-Words (BoW) representation combined with a Multinomial Naive Bayes classifier. In this approach, the textual content of each post and comment is vectorized using raw term frequencies via scikit-learn's `CountVectorizer`, configured with the following parameters: English stop word removal, a maximum vocabulary size of 5,000 features, a minimum document frequency of 2, and a maximum document frequency of 0.95 to filter out near-ubiquitous terms.

The classifier is trained on individual content items (posts and comments) rather than pre-aggregated user documents. User-level aggregation is described in Section III-C4. Multinomial Naive Bayes is chosen for its computational efficiency, its well-documented performance on sparse text representations and its probabilistic output, which enables threshold tuning at the aggregation stage.

Four experimental variants are run, differing in the subreddits included and the minimum content filters applied:

- **BOW_all_sources**. Includes users from all seven subreddits, with no words and content filters applied.
- **BOW_all_sources_filtered**. Like the previous, but with a minimum number of 19 words per post or comment and 24 texts per user.
- **BOW_removed_off_topic**. Excludes users originating from *r/self* and *r/CasualConversation*, with no words and content filters applied.
- **BOW_removed_off_topic_filtered**. Like the previous, but with a minimum number of 29 words per post and comment and 25 texts per user.

The method for determining the values for the minimum number of words and texts in the filtered variants is described in Appendix B.

2) Fine-Tuning Pre-Trained Language Models: The second classification strategy leverages contextual representations from transformer-based language models. We fine-tune `bert-base-uncased` on the ADHD classification task using the Hugging Face `transformers` library, maintaining the same train/validation/test split procedure and user-level evaluation protocol as the BoW baseline.

Each content item is tokenized using the BERT tokenizer with truncation enabled to handle sequences exceeding the model's maximum input length of 512 tokens. No explicit padding to a fixed length is applied during tokenization; instead, dynamic padding is performed at batch creation time minimizing unnecessary computation. The labels are binary integers (1 for ADHD, 0 for control), derived from the user-level self-disclosure tags. ADHD-flagging phrases are removed from the text prior to training, as in the BoW baseline, to prevent the model from learning trivial lexical cues.

The model is initialized from `bert-base-uncased` with a two-label classification head. Fine-tuning is conducted over 2 epochs with a learning rate of 2e-5, a weight decay of 0.01, and a per-device batch size of 32. Mixed-precision training (`fp16=True`) is enabled to reduce memory usage and accelerate computation. The model that achieves the highest validation F1 score at the end of any epoch is retained.

Four BERT fine-tuning experiments are conducted, varying the subreddit sources included and the minimum content filters applied to users:

- **BERT_all_sources**. Includes users from all seven subreddits, with no minimum words and content filters.
- **BERT_all_sources_filtered**. Like the previous, but with a minimum of 29 words per post and comment and 17 texts per user.
- **BERT_removed_off_topic**. Excludes users originating from *r/self* and *r/CasualConversation*, with no words and content filters.
- **BERT_removed_off_topic_filtered**. Like the previous, but with a minimum of 29 words per post and comment and 16 texts per user.

The filter tuning procedure for the filtered variants is described in Appendix B.

3) Multimodal Classification: The third classification strategy moves beyond text-only approaches by leveraging a large multimodal language model capable of jointly reasoning over text and images. We employ LLaVA-NeXT [22], specifically the `llava-v1.6-mistral-7b-hf` variant, which integrates a vision encoder with the Mistral-7B language model [23]. Unlike the BoW and BERT approaches, which require task-specific training, LLaVA is used in a zero-shot configuration: the model is prompted to classify each content item without any fine-tuning on the ADHD corpus.

Four zero-shot prompting variants are evaluated, varying in output format and modality usage:

- **zs_text_only**. The model is given the content text and instructed to respond with only the label `ADHD` or `NON-ADHD`.
- **zs_with_media**. Same prompt format, but accompanied by images associated with the content item.
- **zs_confidence_text_only**. The model is instructed to respond with a JSON object containing both the predicted label and a confidence score between 0.0 and 1.0.

- **zs_confidence_with_media**. Same JSON prompt format, with images included.

The JSON-based prompting strategies are designed to extract a more nuanced confidence signal from the model. When images are available, up to three of them are loaded per datapoint and presented alongside the text in the multimodal input format expected by LLaVA.

Inference is performed by greedy decoding and a maximum of 16 new tokens. For short-format responses, the model directly generates a label (1 for ADHD, 0 for non-ADHD). For JSON-format responses, the label and confidence score are extracted via regular expression; malformed outputs are discarded.

The four prompting variants allow us to assess two orthogonal factors: (1) whether multimodal inputs (text + images) improve classification over text alone, and (2) whether confidence-weighted scoring provides more reliable user-level predictions than hard binary labels.

4) Evaluation Protocol: All models are evaluated under a consistent protocol to ensure fair comparison across classification strategies. For all models, a fixed 70/15/15 train/validation/test split is employed, stratified by ADHD label. For the multimodal (LLaVA) experiments, since there is not a training step, only the validation and tests sets are used. In all cases, splits are performed at the user level to prevent data leakage between partitions.

Predictions are first generated at the post or comment level. For BoW and BERT, each post or comment receives a predicted probability of the positive (ADHD) class. These probabilities are then aggregated to the user level by computing the mean across all items belonging to a given user. For the LLaVA-based experiments, aggregation follows a confidence-weighted scheme: each content item is assigned a label $l_i \in \{0, 1\}$ and a confidence score $c_i \in [0, 1]$ (0 for non-ADHD, 1 for ADHD) and the user-level ADHD score is computed as:

$$S_u = \frac{\sum_{i=1}^{n} l_i \cdot c_i}{\sum_{i=1}^{n} c_i} \tag{1}$$

where n is the number of content items for user u. For the hard-label variants, the same formula is applied with $c_i = 1.0$ for all items, reducing to the proportion of items predicted as ADHD.

For all variants, an optimal decision threshold is determined by sweeping from 0.10 to 0.90 in increments of 0.05 on the validation set and selecting the value that maximizes user-level F1 score. This threshold is subsequently applied to the set to produce final user-level predictions.

We report accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC), all computed at the user level after aggregation. All experiments use a fixed random seed to ensure reproducibility.

IV. Dataset Analysis

A. Dataset Structure

The constructed corpus consists of three main components: a user dataset, a content dataset, and an image directory.

1) User Dataset: The user dataset contains one record per collected user, identified by a 16-character anonymized hash. Each entry includes the user's ADHD label (`has_ADHD`, a boolean indicating whether a self-disclosure phrase was detected in any of their content), the subreddit in which the user was first encountered (`first_found_in`), the total number of posts and comment collected for that user (`n_posts`, `n_comments` and `n_content`), and aggregated statistics on content length, including mean, median, and standard deviation of both word and character counts, computed separately for posts and comments. This dataset serves as the primary source for user-level classification labels and filtering criteria.

2) Content Dataset: The content dataset contains one record per individual post or comment. Each entry is linked to its author via the user id and includes the text body, the content type (post or comment), the subreddit of origin, a UTC timestamp, a flag indicating whether an ADHD disclosure phrase was detected (`has_ADHD_pattern`), and data such as word and character counts or whether the content item has image data (`has_media`). This dataset constitutes the primary input for the text-based classification experiments.

3) Image Directory: The image directory stores all media files downloaded during crawling. Images are organized into sub-directories named by the content id, allowing each image to be linked to its corresponding post via the content dataset.

4) Data Example and Usage: Figure 1 illustrates the structure and contents of the corpus for a single user, as stated in the previous sections. When images are attached to a post or comment, they are stored and implicitly linked to the corresponding item through the directory system. From this data, different levels of analysis are applied: (1) the textual content of posts and comments is processed through the classification pipeline described in Section III-C, to produce content-level predictions; (2) attached images are fed to the multimodal model alongside their corresponding text; and (3) aggregated features are derived to characterize user behavior—as said in Section III-B3—and serve as filtering criteria (view Appendix B).

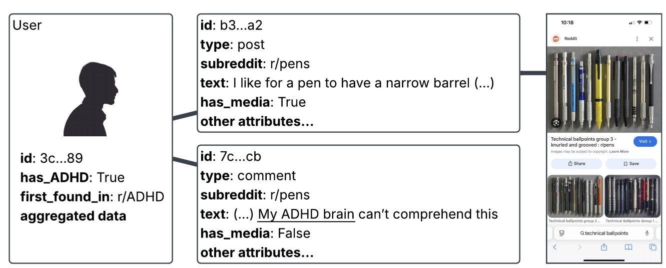

Fig. 1. Schematic overview of a single user's data within the corpus.

B. General Statistics

The corpus comprises 6,847 unique users and 1,994,787 individual content items (posts and comments), alongside 15,770 downloaded images.

Of the 6,847 total users, 1,544 (22.55%) were labeled as having ADHD based on self-disclosure phrases, while 5,303

(77.45%) formed the control group. The class imbalance is addressed during model training through stratified splitting.

Figure 2 presents the distribution of ADHD and control users by the subreddit in which they were first encountered. As expected, ADHD-focused subreddits (*r/ADHD*, *r/adhdwomen*, *r/adhdmeme*) contribute the majority of ADHD-labeled users, while the general-interest and mental health subreddits are predominantly composed of control users.

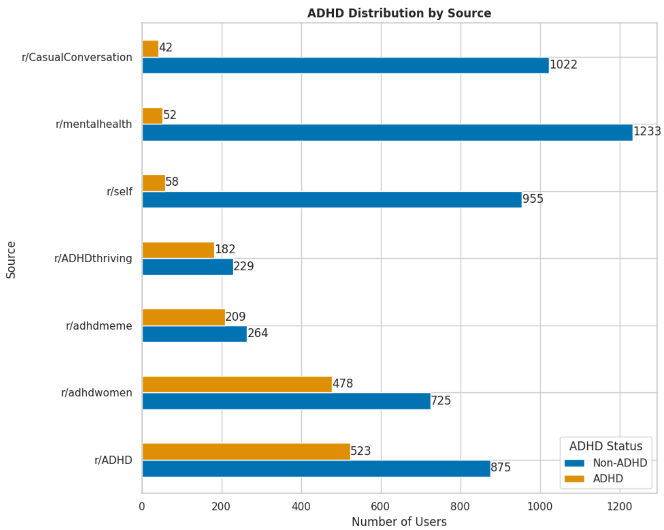

Fig. 2. Distribution of ADHD and control users by the subreddit of first encounter.

Of the total content items, only 7,319 (0,367%) contains an ADHD self-disclosure phrase. These items are retained for corpus analysis but are excluded during model training to prevent the classifier from learning trivial lexical cues.

Users with ADHD contributed 101,010 posts and 1,002,040 comments (1,103,150 total), while control users contributed 92,779 posts and 798,958 comments (891,737 total). The ADHD group produced more content overall (55.30%), despite only forming 22.55% of the dataset, a pattern consistent with previous studies [19].

The ten subreddits with the highest content volume are shown in Figure 3. Note that these subreddits represent the source of the content retrieved from the users' posting history, not the original subreddits that were searched to identify the users.

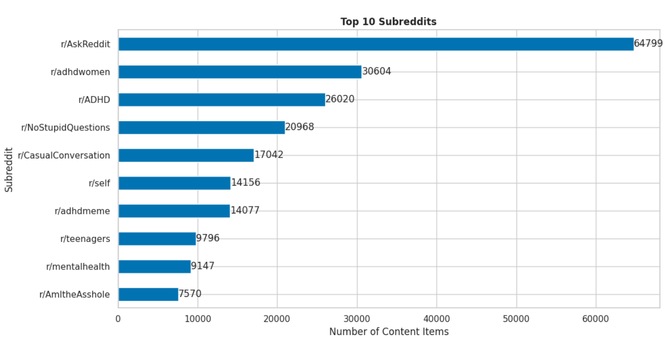

Fig. 3. The ten subreddits with the highest content volume.

C. Text Counts

Figures 4 and 5 present blox-plots of the number of posts (without outliers) and comments per user, stratified by ADHD status. The median user with ADHD published 22.5 posts and 344.0 comments, compared to 2.0 posts and 11.0 comments for control users. ADHD users exhibit a higher median count in both types of content, suggesting a greater tendency towards interactive, conversational engagement.

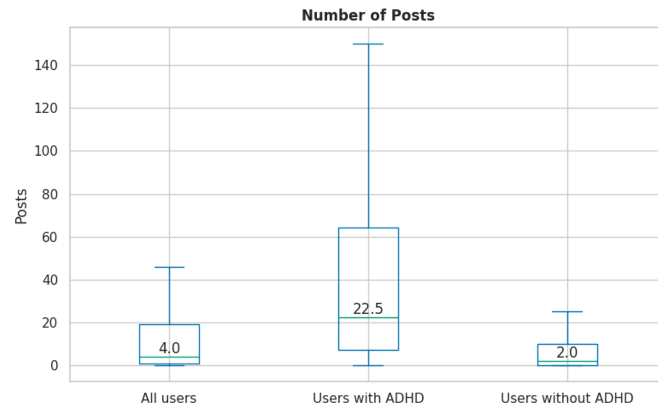

Fig. 4. Number of posts per user (without outliers), stratified by ADHD status.

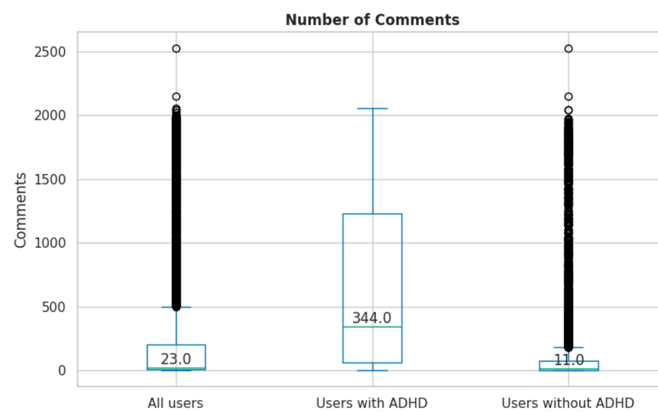

Fig. 5. Number of comments per user, stratified by ADHD status.

Figures 6 and 7 display the distribution of median word counts per user for posts and comments (both without outliers), respectively. For posts, the median word count per user is 90.0 and 89.5 for the control group. For comments, the corresponding figures are 23.0 and 19.0. Character count distributions, shown in Figures 8 and 9, follow a similar pattern. Posts and comments lengths are comparable between groups.

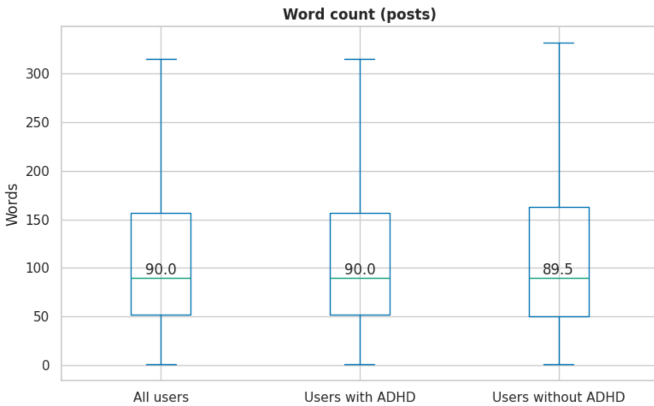

Fig. 6. Distribution of median word counts per user for posts (without outliers), stratified by ADHD status.

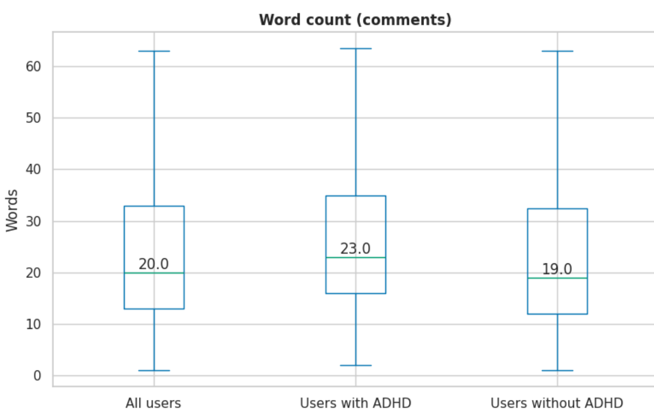

Fig. 7. Distribution of median word counts per user for comments (without outliers), stratified by ADHD status.

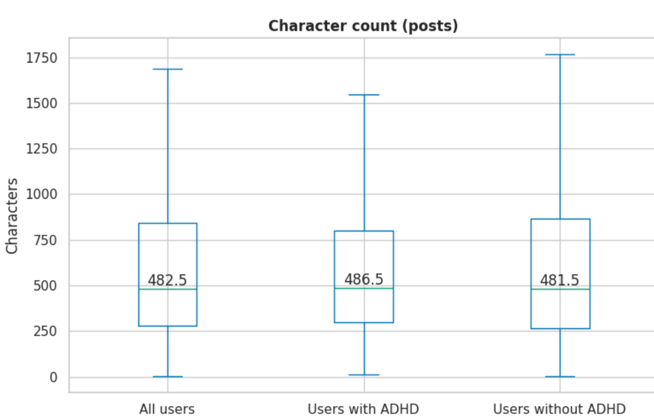

Fig. 8. Distribution of median character counts per user for posts (without outliers), stratified by ADHD status.

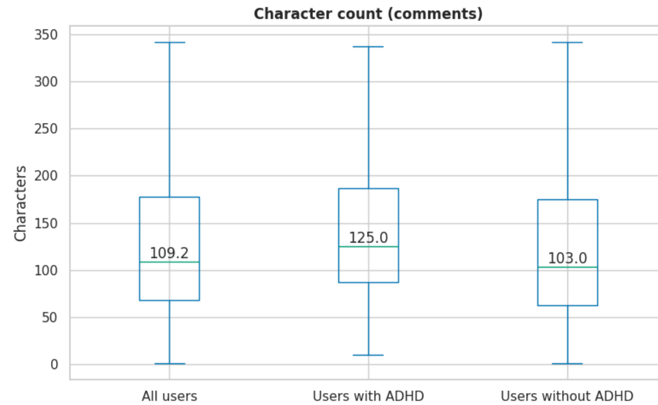

Fig. 9. Distribution of median character counts per user for comments (without outliers), stratified by ADHD status.

D. Content Analysis

This subsection explores the semantic and affective content of the corpus through emotion, sentiment and theme analyses. All text samples are drawn from a stratified random sample of 5,000 items per group to ensure computational tractability.

Emotion probabilities were extracted using a model[3] which classifies text into seven emotion categories: anger, disgust, fear, joy, neutral, sadness, and surprise. Figure 10 compares the median emotion probabilities between the ADHD an control groups, excluding the neutral category to highlight differences among affective emotions. Surprise is the most prevalent emotion for both groups, followed by disgust, anger, sadness, joy, and fear. ADHD users exhibit higher median scores for the negative emotions (sadness, anger, fear and disgust), compared to controls.

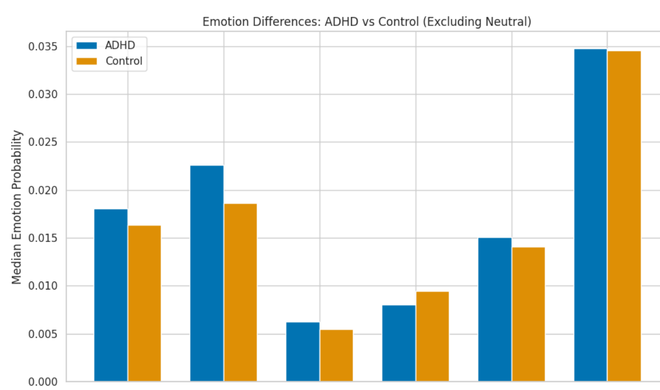

Fig. 10. Median emotion probabilities excluding neutral.

Sentiment probabilities (positive, negative, neutral) were also obtained using a model[4]. Figure 11 compares sentiment distributions between groups, excluding the neutral category. Negative sentiment is the most prevalent for both groups. The ADHD group shows a higher proportion of negative sentiment

[3]The model `j-hartmann/emotion-english-distilroberta-base` was used [20]

[4]The model `j-hartmann/sentiment-roberta-large-english-3-classes` was used [21]

and a slightly lower positive sentiment relative to the control group.

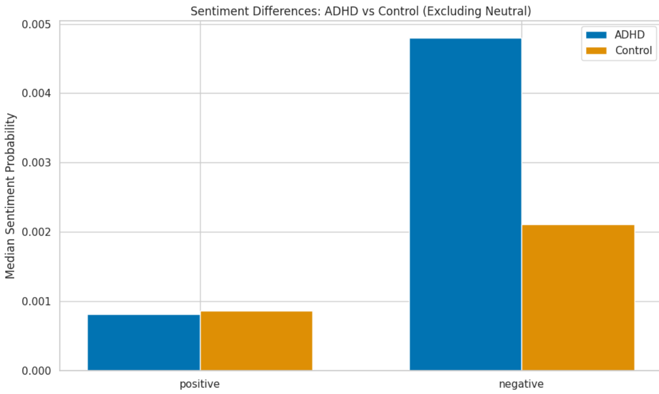

Fig. 11. Median sentiment probabilities excluding neutral.

Thematic content was analyzed using the Empath lexicon, which provides normalized frequencies for 194 pre-defined semantic categories. Figures 12 and 13 display the top 10 themes for each group. Overall, both groups exhibit similar thematic distributions, suggesting an overlap in general discourse domains. Core themes such as "negative emotion," "positive emotion," "speaking," "communication," and "friends" appear among the most frequent categories in both populations, indicating that everyday social and affective language dominates across groups.

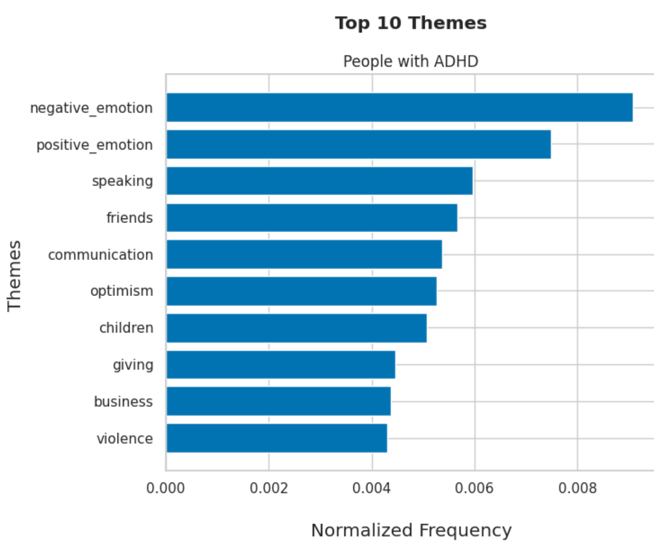

Fig. 12. Top 10 Empath themes for the ADHD group.

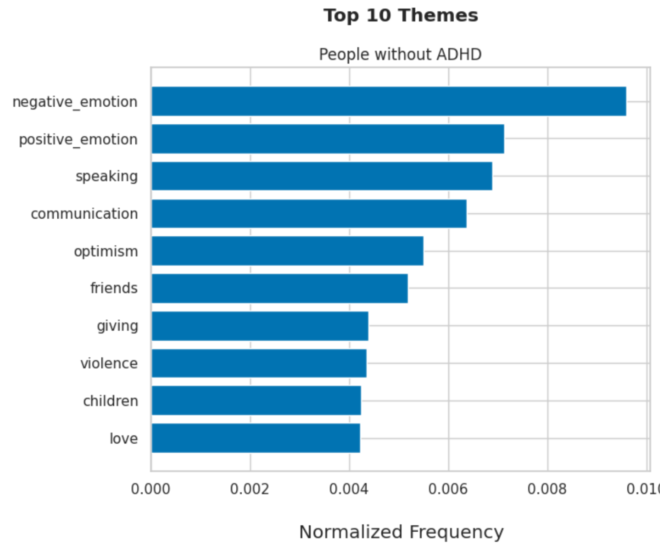

Fig. 13. Top 10 Empath themes for the control group.

E. Temporal Analysis

Beyond static linguistic features, the corpus enables the study of user behavior over time. In this subsection, we analyze the temporal distribution of content, posting frequency, and engagement patterns to assess whether ADHD and control users differ in how and when they contribute to the platform.

Figure 14 presents a stacked area chart of total posts and comments per year. Content volume peaks in 2025, which reflects the data collection methodology rather than an organic trend: although the crawler retrieved the full posting history of each user, users were initially discovered by their most recent activity in the tracked subreddits during the crawling period (May of 2026). This sampling strategy naturally biases the corpus toward users who were active at the time of collection, resulting in an over-representation of content from the most recent full year.

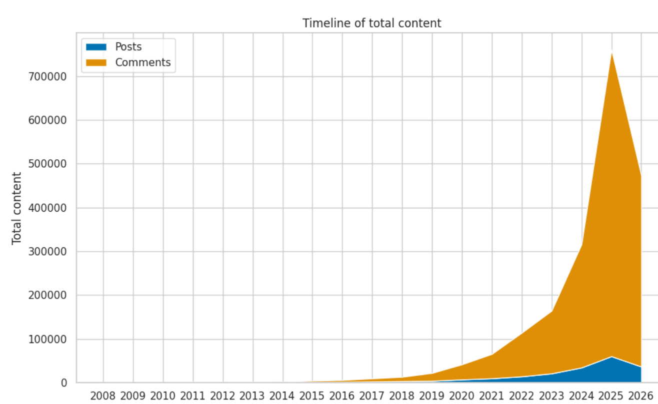

Fig. 14. Stacked area chart of total posts and comments per year.

Figure 15 plots the median number of content items per user per year, stratifies by ADHD status. ADHD users tend to produce more content per year than control user, with the biggest gap in 2025.

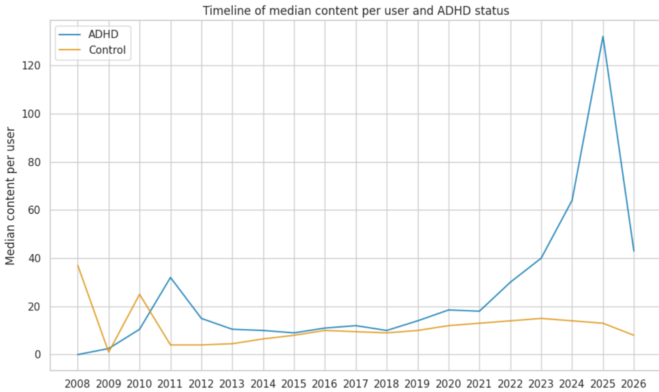

Fig. 15. Median number of content items per user per year, stratified by ADHD status.

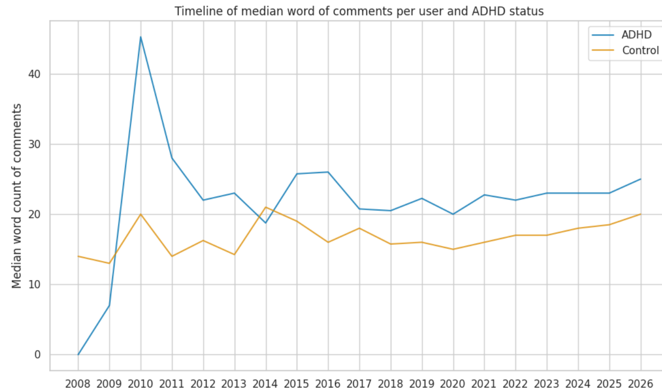

Fig. 17. Median word count of comments per user per year, stratified by ADHD status.

Figures 16 and 17 show the median word count of posts and comments per user per year. Both post and comment lengths have remained relatively stable over time for both groups since 2011. Figure 18 presents the proportion of active days (days with at least one post or comment) within each user's recorded date range. The median proportion is comparable between ADHD and control users, indicating that both groups are equally consistent in their engagement over time. However, the third quartile is notably higher for control users, suggesting that the most active control users sustain their engagement across a larger fraction of their observed lifespan. Conversely, the ADHD group exhibits a wider spread of outliers above the third quartile, indicating the presence of a small subset of highly active individuals whose posting frequency approaches daily consistency.

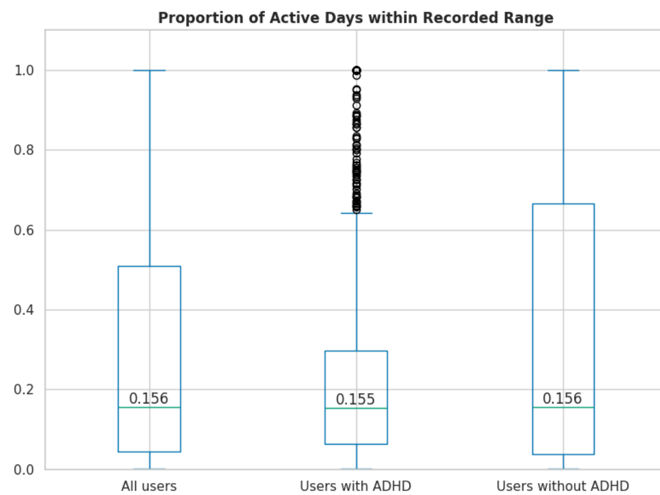

Fig. 18. Proportion of active days (days with at least one post or comment) within each user's recorded date range, stratified by ADHD status.

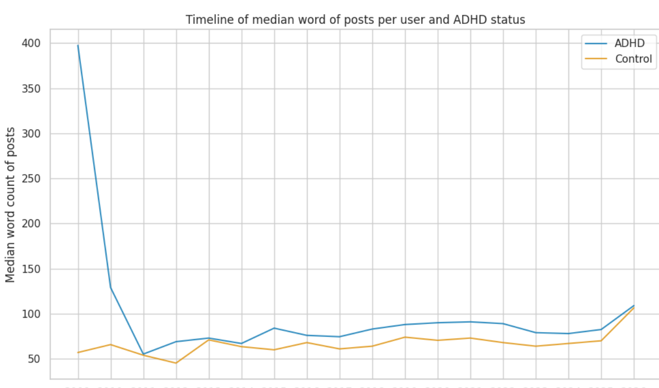

Fig. 16. Median word count of posts per user per year, stratified by ADHD status.

Figures 19 and 20 display the median daily content published per user. The median ADHD user publishes 2.0 items per active day, twice the median of 1.0 for control users. This suggests that, on days when they engage with the platform, ADHD users tend to post or comment more frequently. However, control users exhibit higher outlier values, indicating that the most prolific daily contributors belong to the control group.

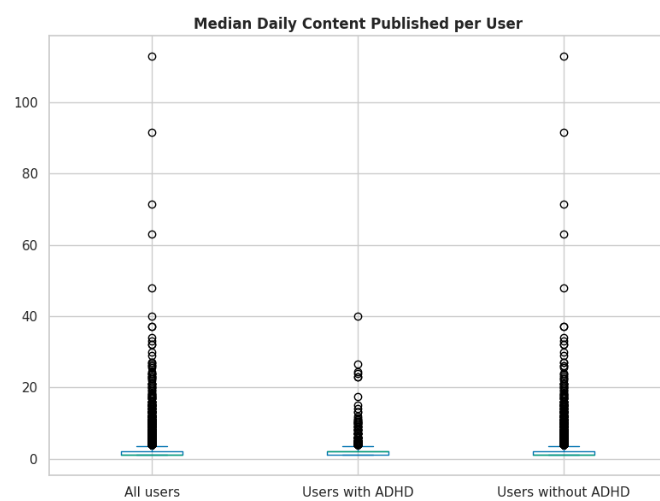

Fig. 19. Median daily content published per user, stratified by ADHD status.

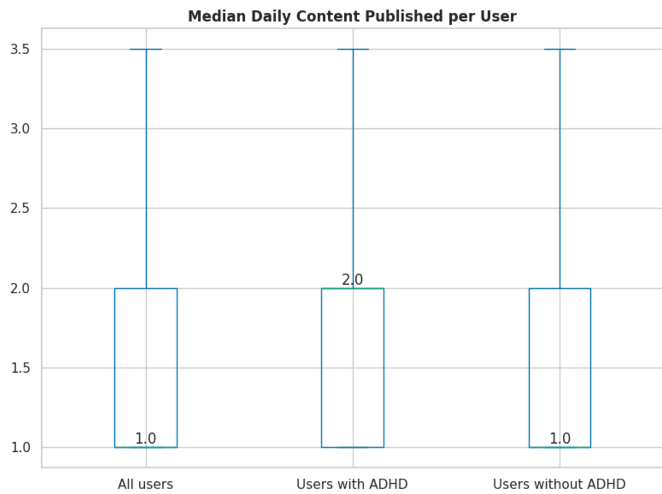

Fig. 20. Median daily content published per user (without outliers), stratified by ADHD status.

F. Media Content Analysis

This subsection characterizes the visual content in the corpus.

Of the 1,994,787 total content items, 15,770 (0.791%) include attached images. This low proportion reflects the primarily text-based nature of the tracked subreddits. Figure 21 ranks the subreddits by the percentage of content containing media (filtered to subreddits with at least 10 total posts). Subreddits such as *r/CRFla* and *r/fussball* exhibit the highest media percentages (both 100%), reflecting their image-centric posting cultures. Note that these subreddits represent the source of the content retrieved from the users' posting history, not the original subreddits that were searched to identify the users.

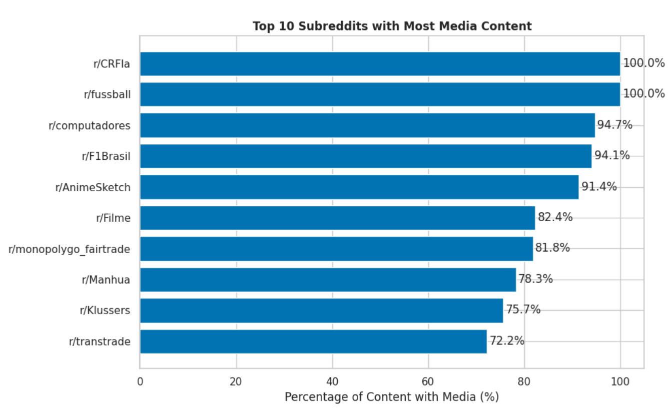

Fig. 21. Top 10 subreddits by percentage of content containing media (minimum 10 total posts).

V. Results

This section presents the results of the classification experiments described in Section III-C. We evaluate and compare three classification strategies of increasing complexity: a Bag-of-Words baseline with Multinomial Naive Bayes, a fine-tuned BERT model, and a zero-shot multimodal approach using LLaVA-NeXT. For all strategies, user-level predictions are obtained by aggregating content-item probabilities and applying an optimal decision threshold, selected to maximize validation F1. We examine both validation and test set performance across all variants, analyze generalization gaps, and investigate the impact of user-level content filters on classification quality. The section concludes with a synthesis of findings across all experimental configurations.

A. Experimental Configuration

All experiments follow the evaluation protocol established in Section III-C4: user-level aggregation of content-item predictions, threshold optimization on the validation set, and reporting of accuracy, precision, recall, F1, and ROC-AUC at the user level.

All experiments were conducted on the High Performance Computing Cluster (HPC) ctcomp3, operated by CiTIUS. Depending on resource availability, jobs were executed on one of four GPU-enabled nodes: hpc-gpu3 and hpc-gpu4, equipped with AMD EPYC 7543 CPUs and NVIDIA A100 GPUs (40 GB and 80 GB, respectively), or hpc-gpu5 and hpc-gpu6, equipped with AMD EPYC 9255 CPUs and NVIDIA L4 GPUs (24 GB). Each experiment used a single GPU.

B. Bag of Words

Figure 22 presents the test set ROC curves for the four BoW variants. All variants perform substantially better than the random classifier baseline. The variants are tightly clustered, indicating that filtering strategies had limited impact on discriminative power. A brief flat region near the origin (zero sensitivity at very low false positive rates) is observed across all curves.

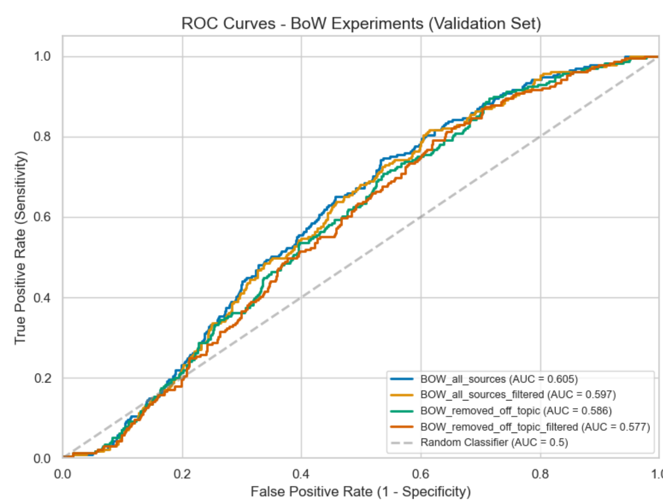

Fig. 22. ROC curves – BoW variants (test set).

Table I reports test set performance. The variant using all sources with filtering (BOW_all_flt) achieved the highest accuracy (0.498) and precision (0.272), while the unfiltered all-sources variant (BOW_all) obtained the best F1 (0.407) and AUC (0.605). Removing off-topic content without filtering

(BOW_off) produced the highest recall (0.926) but the lowest accuracy (0.371) and precision (0.252), indicating a strong bias towards positive predictions at the expense of false positives. Contrary to our expectations, retaining all content did not substantially improve discriminative performance, despite our hypothesis that potential off-topic users would provide a clearer contrast with the ADHD group.

TABLE I
TEST METRICS FOR BAG-OF-WORDS VARIANTS.

Experiment	Thr.	Acc.	F1	Prec.	Rec.	AUC
BOW_all	0.50	0.467	**0.407**	0.271	0.817	**0.605**
BOW_all_flt	0.55	**0.498**	0.398	**0.272**	0.742	0.597
BOW_off	0.55	0.371	0.397	0.252	**0.926**	0.586
BOW_off_flt	0.65	0.454	0.399	0.265	0.812	0.577

Thr. = threshold; Acc. = accuracy; Prec. = precision; Rec. = recall; AUC = ROC-AUC. all = all sources; off = removed off-topic; flt = filtering applied.

C. Fine-Tuned BERT

Figure 23 presents the test set ROC curves for the BERT variants. All variants perform substantially better than the random classifier baseline. Notably, the two variants using all sources achieve superior performance, particularly in the middle range of the ROC curve.

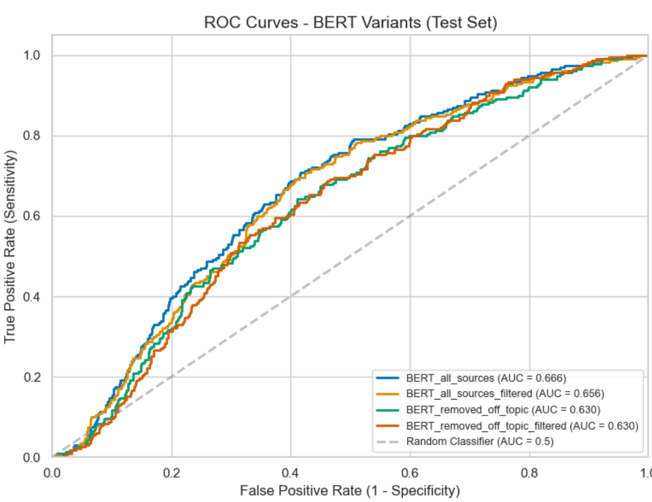

Fig. 23. ROC curves – BERT variants (test set).

Table II reports test set performance. The unfiltered all-sources variant (BERT_all) achieved the highest accuracy (0.626), precision (0.330), and AUC (0.666), and tied for the best F1 (0.439) with its filtered counterpart (BERT_all_flt). The variant removing off-topic content without filtering (BERT_off) produced the highest recall (0.800) but the lowest accuracy (0.485) and precision (0.276), mirroring the precision–recall trade-off observed in the BoW experiments. Filtering slightly improved the off-topic variant (BERT_off_flt), raising accuracy to 0.528 while decreasing recall to 0.743, though AUC remained unchanged at 0.630. Overall, the all-sources variants outperformed their off-topic counterparts across most metrics.

TABLE II
TEST METRICS FOR BERT VARIANTS.

Experiment	Thr.	Acc.	F1	Prec.	Rec.	AUC
BERT_all	0.55	**0.626**	**0.439**	**0.330**	0.652	**0.666**
BERT_all_flt	0.55	0.571	**0.439**	0.310	0.748	0.656
BERT_off	0.60	0.485	0.411	0.276	**0.800**	0.630
BERT_off_flt	0.65	0.528	0.414	0.287	0.743	0.630

Thr. = threshold; Acc. = accuracy; Prec. = precision; Rec. = recall; AUC = ROC-AUC. all = all sources; off = removed off-topic; flt = filtering applied.

D. Multimodal Classification

Figure 24 presents the test set ROC curve for the multimodal variant. Two observations emerge from these results. First, the addition of multimodal content does not meaningfully affect AUC values: text-only and multimodal variants differ by at most 0.001 across all configurations. Second, the confidence-weighted variants appear to produce only two distinct probability values. As shown in Figure 25, the aggregated user-level ADHD scores are limited to 0.5 and 1.0. This suggests that the model either fails to generate probabilities across the full [0,1] range, or outputs improperly formatted JSON that the parsing code cannot interpret. When no valid confidence score is extracted—resulting in a confidence sum of zero—the system defaults to 0.5; conversely, when the computed score exceeds 1.0, it is capped at 1.0. As a result, the final user-level scores collapse to only these two values. Addressing this issue may require a more capable model or improved prompt engineering.

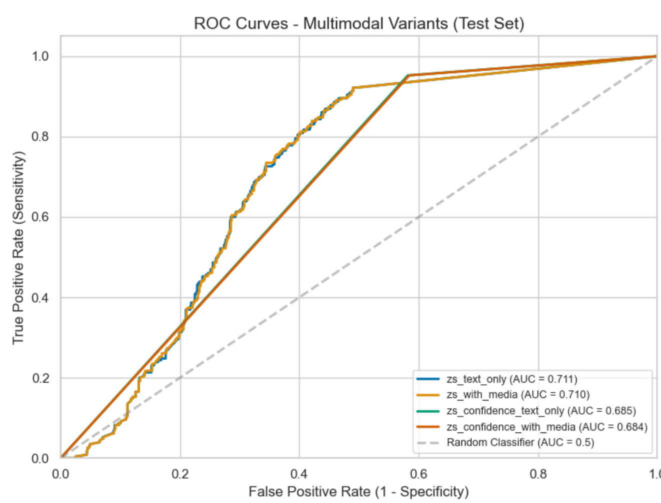

Fig. 24. ROC curve – Multimodal variant (test set).

Table III reports test set performance. The hard-label variants (zs_text and zs_media) achieved the highest accuracy (0.706) and AUC (0.711 and 0.710, respectively), but at the cost of very low recall (0.213 and 0.217) and F1 (0.245 and 0.249). In contrast, the confidence-weighted variants (zs_conf_text and zs_conf_media) yielded substantially higher F1 (0.480 and 0.479) and near-perfect recall (0.952 for both), though with lower accuracy (0.538 and 0.536) and AUC (0.685 and 0.684). This divergence reflects a clear labeling bias: the hard-label variants defaulted heavily toward negative

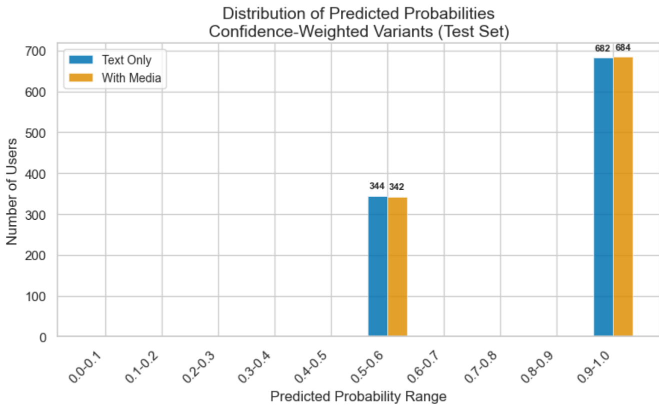

Fig. 25. Distribution of aggregated user-lebel ADHD scores for confidence-weighted multimodal variants (test set).

predictions, while the confidence-weighted variants exhibited the opposite tendency. This discrepancy can be clearly seen in the confusion matrices shown in Figure 26. The negligible difference between text-only and multimodal configurations across all metrics confirms that the visual modality contributed little to the model's decision, regardless of the scoring method.

TABLE III
TEST METRICS FOR MULTIMODAL VARIANTS.

Experiment	**Thr.**	**Acc.**	**F1**	**Prec.**	**Rec.**	**AUC**
zs_text	0.10	**0.706**	0.245	0.288	0.213	**0.711**
zs_media	0.10	**0.706**	0.249	0.291	0.217	0.710
zs_conf_text	0.50	0.538	**0.480**	**0.321**	**0.952**	0.685
zs_conf_media	0.50	0.536	0.479	0.320	**0.952**	0.684

Thr. = threshold; Acc. = accuracy; Prec. = precision; Rec. = recall; AUC = ROC-AUC. zs = zero-shot; conf = confidence-weighted; text = text only; media = text + images.

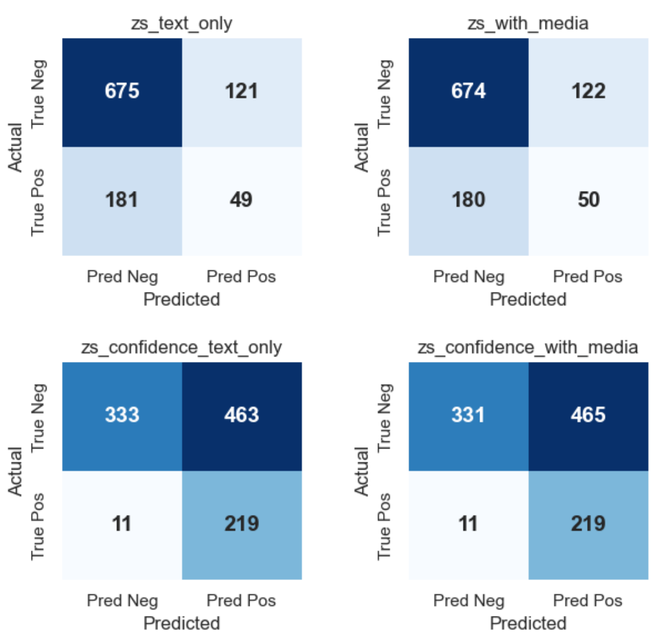

Fig. 26. Test set confusion matrices for multimodal variants.

E. All Variants Comparison

Figure 27 overlays the test ROC curves for all variants, with each strategy group (BoW, BERT, and multimodal) shown in a distinct color to facilitate cross-family comparison. A clear performance hierarchy emerges: multimodal variants achieve the highest AUC values, followed by BERT, and then BoW. However, the multimodal group exhibits greater internal variability: the confidence-weighted variants display distinctively shaped curves due to their binary probability outputs, yet still attain AUC scores exceeding those of the best BERT variants. Despite this irregularity, the overall trend confirms that increasing model complexity yields consistent improvements in discriminative power.

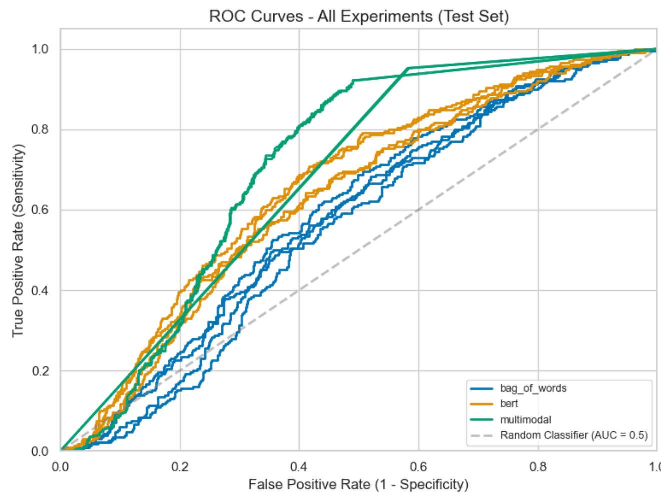

Fig. 27. ROC curves – All variants (test set).

Table IV compares test metrics across all experimental configurations. A trend emerges with respect to model complexity, as was seen in the previous figure: BoW variants achieve the lowest AUC values (0.554–0.599), BERT variants occupy the middle range (0.630–0.666), and multimodal variants reach the highest (0.684–0.711). However, despite this consistent improvement in AUC, overall performance remains modest: no variant exceeds an AUC of 0.711, indicating that even the best models struggle to reliably separate the two classes. Moreover, the relatively higher F1 scores achieved by the confidence-weighted multimodal variants (0.479–0.480) come at the cost of near-perfect recall (0.952) but low precision (0.320–0.321), reflecting a strong bias toward positive predictions. This scoring strategy is further limited by the model's inability to produce fine-grained confidence estimates, producing only two values (0.5 and 1.0). Adding visual information did not meaningfully alter any metric across either strategy. Taken together, these results underscore the difficulty of the task and suggest that none of the baseline approaches tested here provides a sufficiently reliable solution.

TABLE IV
TEST METRICS FOR ALL VARIANTS.

Experiment	Thr.	Acc.	F1	Prec.	Rec.	AUC
BOW_all	0.50	0.460	0.400	0.267	0.804	0.599
BOW_all_flt	0.55	0.491	0.390	0.267	0.726	0.583
BOW_off	0.55	0.354	0.391	0.248	0.926	0.573
BOW_off_flt	0.65	0.426	0.377	0.249	0.774	0.554
BERT_all	0.55	0.626	0.439	**0.330**	0.652	0.666
BERT_all_flt	0.55	0.571	0.439	0.310	0.748	0.656
BERT_off	0.60	0.485	0.411	0.276	0.800	0.630
BERT_off_flt	0.65	0.528	0.414	0.287	0.743	0.630
zs_text	0.10	**0.706**	0.245	0.288	0.213	**0.711**
zs_media	0.10	**0.706**	0.249	0.291	0.217	0.710
zs_conf_text	0.50	0.538	**0.480**	0.321	**0.952**	0.685
zs_conf_media	0.50	0.536	0.479	0.320	**0.952**	0.684

Thr. = threshold; Acc. = accuracy; Prec. = precision; Rec. = recall; AUC = ROC-AUC. all = all sources; off = removed off-topic; flt = filtered; zs = zero-shot; conf = confidence-weighted; text = text only; media = text + images.

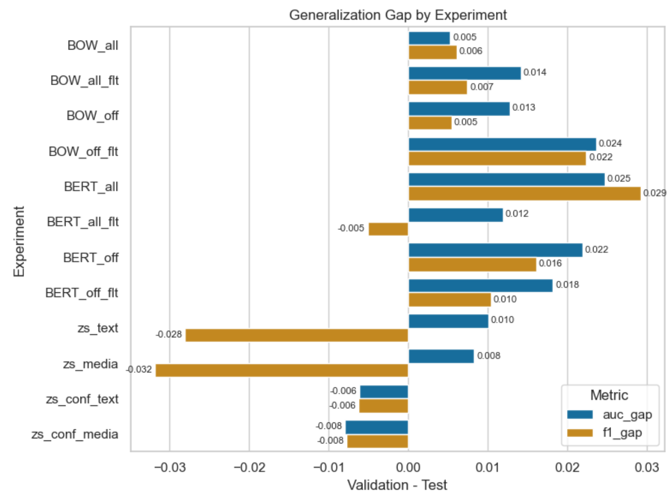

Fig. 28. Generalization gap (validation − test) for ROC-AUC and F1 across all variants.

F. *Overfitting Analysis*

To assess generalization, we compare validation and test performance for each variant. Figure 28 shows the generalization gap (validation minus test) for ROC-AUC and F1 across all experiments. Positive values indicates higher validation than test performance, a sign of potential overfitting. Overall, the gaps are small—most below 0.03—indicating that overfitting is not a serious concern across any strategy. However, several patterns are worth noting. Within the BoW family, the filtered off-topic variant (BOW_off_flt) shows the largest AUC gap (0.024) and F1 gap (0.022), suggesting that the combination of content filtering and thresholding on the already small off-topic subset may have led to slight overtunning on the validation set. BERT variants exhibit slightly larger AUC gaps (0.012–0.025) than BoW, with BERT_all showing the largest F1 gap (0.029).

Interestingly, the multimodal variants display a contrasting pattern by scoring method. The hard-lebel variants (zs_text adn zs_media) show small positive AUC gaps (0.008–0.010), but negative F1 gaps (-0.028 and -0.032), meaning test F1 was actually higher than validation F1. The confidence-weighted variants (zs_conf_text and zs_conf_media) show negative gaps for both AUC and F1 (-0.006 to -0.008), indicating marginally better test than validation performance. These negative gaps, while unusual, are likely an artifact of the model's restricted probability outputs (0.5 and 1.0) and the small absolute differences involved, rather than evidence of genuine improvement on unseen data. Overall, the consistently small gaps across all variants indicate that the threshold selection procedure did not introduce meaningful overfitting; rather, the modest performance reflects the inherent difficulty of the task, which persists even when models generalize well.

VI. ERROR ANALYSIS

In this section, we examined whether incorrectly classified content items and users exhibited differences in word count or content volume compared to correctly classified ones. This analysis is applied to the following variants: *BOW_all_sources*, *BOW_removed_off_topic*, *BERT_all_sources* and *BERT_removed_off_topic*. Multimodal variants are excluded, as they were not trained and therefore have no training set to filter.

Figure 29 presents the distribution of content volume for correctly and incorrectly classified users for one representative variant (*BERT_all_sources*); the remaining figures are provided in Appendix C. Incorrectly classified users have noticeably lower median content counts than their correctly classified counterparts. This indicates that users with very few posts or comments are harder to classify reliably, likely because the aggregated signal is based on insufficient evidence. This observation supports the introduction of a minimum content filter, which excludes users below a certain content threshold during training (see Appendix B for the full threshold selection procedure).

Word count differences were also examined. However, the median word counts are nearly identical across groups, indicating that content length alone is not a strong predictor of classification error at the item level.

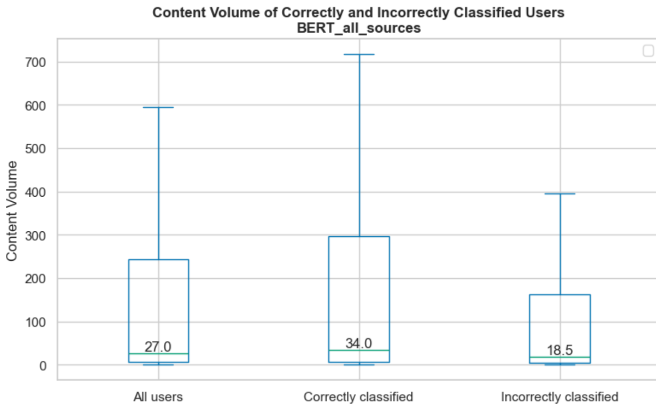

Fig. 29. Content volume (number of posts and comments) of correctly and incorrectly classified users for the *BERT_all_sources* validation set (outliers excluded).

VII. Discussion

This study evaluated three classification strategies for detecting ADHD from social media text. The results reveal a consistent performance hierarchy: BoW (AUC 0.554–0.599) < BERT (AUC 0.630–0.666) < multimodal (AUC 0.684–0.711). This progression confirms that increasing model complexity yields improvements in discriminative power, though the gains from BERT to multimodal are modest and come with important caveats. Critically, none of the tested strategies achieved strong performance: the best F1 across all experiments is 0.480, and precision never exceeds 0.330. These baselines thus serve primarily to illustrate the value of the corpus as a benchmark and to demonstrate the considerable difficulty of the task.

The BoW baseline, despite its simplicity, achieved reasonable recall (up to 0.926) and provided an interpretable feature space. However, its low precision (0.248–0.267) indicates that lexical features alone are insufficient to reliably distinguish ADHD from control users. The finding that including off-topic content did not improve performance runs counter to our initial expectation. We hypothesized that off-topic users (predominantly controls) would provide a clearer contrast with the ADHD group; instead, the relevant linguistic signal appears distributed across all content domains. This suggests that the markers of ADHD in social media language are not confined to disorder-specific vocabulary or topics, but rather permeate everyday communication.

BERT fine-tuning yielded the most balanced results, with the best overall precision (0.330) and one of the highest AUC among text-only models (0.666). The modest absolute performance (F1 $\leq$ 0.439), however, underscores the inherent difficulty of the task: ADHD-related linguistic patterns are subtle and overlap considerably with normal variation in online communication.

The multimodal results present a more complex picture. While the zero-shot LLaVA-NeXT achieved the highest overall AUC (0.711), this was driven by the hard-label variants, which produced extremely low recall (0.213–0.217) and F1 (0.245–0.249), effectively defaulting to negative predictions for most users. The confidence-weighted variants, in contrast, achieved the best F1 (0.480) and near-perfect recall (0.952) but at the cost of accuracy and a strong bias toward positive predictions. The observation that the model produced only two score values (0.5 and 1.0) reveals a limitation: the zero-shot prompting strategy failed to elicit well-calibrated confidence estimates. Rather than reflecting genuine multimodal reasoning, the model's outputs appear to be dominated by the texts, with images contributing negligibly to the final decision. Future work with more capable models, improved prompt engineering, or domain-specific multimodal corpus may unlock the potential of visual media for this task.

In summary, the consistently modest results across all three strategies—from simple lexical models to state-of-the-art multimodal systems—underscore that ADHD detection from social media remains an open and challenging problem. The baselines reported here establish a reference point for future work using this corpus and highlight the need for approaches specifically designed to capture the subtle, distributed linguistic markers associated with ADHD.

The modest performance observed across all strategies (F1 $\leq$ 0.480, AUC $\leq$ 0.711) is consistent with findings from the eRisk shared tasks, where F1 scores typically fall below 0.70 for mental health detection from social media [12].

Methodologically, this study extends prior work in several ways. Unlike Coppersmith et al. [11], who classified ADHD among nine other conditions using Twitter data, our work focuses exclusively on ADHD with a richer per-user content sample drawn from Reddit, enabling more detailed linguistic analysis. The use of self-reported diagnosis via pattern matching follows established practice in computational mental health [16], though it introduces validity concerns discussed below.

A. Limitations and Practical Implications

Several limitations warrant discussion. First, the validity of self-reported ADHD diagnosis on Reddit cannot be verified. While pattern-based labeling has been widely used in prior work [16], it is susceptible to both false positives (users who mention ADHD without having the condition) and false negatives (users with ADHD who never explicitly disclose it). This labeling noise likely contributes to the modest classification performance and may obscure genuine linguistic markers.

Second, Reddit's demographic composition introduces bias. The platform skews male, younger and predominantly United States-based [14], and our inclusion of *r/adhdwomen* only partially mitigates gender imbalance. ADHD presentation varies by gender [13], and a more representative sample would be needed to capture this diversity. Furthermore, ADHD diagnosis rates and clinical practices vary substantially across countries, reflecting differences in diagnostic criteria, cultural attitudes toward mental health, and healthcare access [24].

Third, the corpus may contain users with comorbid conditions (e.g., depression, anxiety), which are highly prevalent among individuals with ADHD [25]. Since our labeling procedure only flags ADHD disclosure, control users with undisclosed mental health conditions may dilute the contrast between groups, and ADHD users with comorbidities may exhibit linguistic patterns driven by multiple conditions simultaneously.

Fourth, computational constraints limited the multimodal experiments. Images were restricted to three per content item, and the choice of LLaVA-NeXT (7B parameters) was dictated by GPU memory availability and time constraints. Larger multimodal models or more sophisticated image processing pipelines may yield different results.

Finally, the temporal bias toward 2025 content (due to the crawling strategy) may over-represent recently active users, potentially excluding individuals whose ADHD-related posting patterns differ from those currently engaged on the platform.

The modest performance achieved by even the best models (F1 = 0.480) indicates that these systems are not suitable for clinical diagnosis. However, they may serve as scalable screening or awareness tools, consistent with the conclusions of the eRisk survey [12]. Crucially, such tools should be designed to support, and not replace, clinical judgment, and their outputs should be interpreted with caution given the limitations of self-reported diagnostic labels and platform-specific demographic biases.

VIII. Conclusions and Future Work

This study set out to address two research questions: (1) *What linguistic features distinguish the social media language of individuals with ADHD from that of neurotypical controls?* and (2) *How do different computational models compare in their ability to classify ADHD users?* Both questions have been addressed through the construction of a novel multimodal corpus, a comparative linguistic analysis, and a systematic benchmarking of three classification paradigms.

Our corpus analysis revealed several distinguishing characteristics of ADHD users on Reddit. Individuals with ADHD produced significantly more content overall (55.30% of total items, despite comprising only 22.55% of users), and they exhibited higher median daily posting frequency (2.0 vs. 1.0 items per active day), consistent with clinical observations of increased verbal output and interactive engagement. Emotion and sentiment analyses revealed elevated negative affect (sadness, anger, fear, disgust) and reduced positive sentiment in the ADHD group, aligning with the well-documented emotional dysregulation associated with the disorder. However, thematic analysis using the Empath lexicon showed substantial overlap between groups, indicating that ADHD-related linguistic differences are distributed across everyday discourse domains rather than concentrated in disorder-specific topics.

The classification experiments demonstrated that while increasing model complexity yields consistent improvements in AUC, from BoW ($\leq$0.599) to BERT ($\leq$0.666) to multimodal ($\leq$0.711), absolute performance remains modest, with F1 scores peaking at 0.480. This reflects the inherent difficulty of the task: ADHD-related linguistic signals are subtle, overlap with normal communicative variation, and are not confined to any single topic or lexical domain. The finding that including off-topic content degraded performance further supports this conclusion.

This work makes three main contributions. First, we introduce a publicly available, anonymized, multimodal corpus of Reddit data from 6,847 users (1,544 ADHD, 5,303 control), comprising nearly 2 million content items and almost 15,000 images (counts reflect the dataset after removal of sexually explicit content for public release, as described in Section III-A4). Second, we establish a multimodal classification benchmark for ADHD detection, evaluating zero-shot LLaVA-NeXT alongside traditional BoW and fine-tuned BERT baselines. Third, we provide a detailed linguistic characterization of ADHD users on social media, identifying both quantitative (posting volume, daily frequency) and affective (elevated negative emotion, reduced positive sentiment) markers that distinguish the group.

Several directions emerge from this study. First, validation against clinically confirmed ADHD diagnoses, rather than self-reported labels, would strengthen the reliability of the identified linguistic markers and the trained models. Second, the development of explainable models is important for clinical adoption. While BERT and LLaVA offer improved performance over BoW, their opacity limits their utility for clinicians seeking to understand why a particular user was flagged. Third, multilingual extension of this work would address an important gap. The current corpus is English-only, yet ADHD is a global condition. Cross-linguistic analysis could reveal whether the linguistic markers identified here are language-specific or reflect universal features of ADHD communication. Finally, as larger and more capable multimodal models become available, revisiting the role of visual information is warranted. While images contributed negligibly in the present study, future models with improved visual reasoning capabilities may extract signals from image content, composition, or aesthetic properties that correlate with ADHD-related behavioral patterns.

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Appendix A
Code Repository

The code and resources developed for this work are publicly available at the following repository:

https://github.com/sotelo-s/ADHD-Reddit-Research

The repository is organized as follows:

- **`code/crawler/`**. Reddit crawler (`crawler.py`) and configuration files (`adhd_phrases.json`, `adhd_search_subreddits.json`).
- **`code/data_processing/`**. Post-processing scripts: `join_files.py`, `data_cleaning.py`, `update_adhd.py`, `media_anonymizer.py`, and `nsfw_censoring.py`.
- **`code/training/`**. Classification experiments: `bag_of_words.py`, `BERT_fine_tuning.py`, and `multimodal_zeroshot.py`.
- **`code/analysis/`**. Jupyter notebooks for data exploration (`corpus_view.ipynb`) and result analysis (`experiment_result.ipynb`), plus compiled metrics (`data/experiment_results.csv`) and per-experiment prediction files (`data/predictions/`).
- **`datasets/`**. The anonymized corpus (user and content CSV files).
- **`Detection_of_Personality_Disorders_based_on_Narratives.pdf`**. The present manuscript.

The media dataset and a CSV file listing which images were transformed as part of the face anonymization process, are available at:

https://nubeusc-my.sharepoint.com/:f:/g/personal/sabrina_sotelo_rai_usc_es/IgCNhdlA_5aKSZy-TEFzY9AkAbEeDloX6NEtYm3xPT-_2J4?e=UJsidT

Appendix B
Filtering Criteria for Classification Variants

This section describes how filtering was applied for the classification variants: *BOW_all_sources_filtered*, *BOW_removed_off_topic_filtered*, *BERT_all_sources_filtered* and *BERT_removed_off_topic_filtered*. Multimodal variants are excluded, as they were not trained and therefore have no training set to filter.

We determined two values for each variant: (1) the minimum number of words a post or comment must contain, and (2) the minimum number of of texts a user must have. These filters are applied exclusively to the training set, ensuring that the validation and test splits remain identical across all variants for direct comparability.

To select these values, we examined the validation results: setting aside outliers, how does the F1 score improve when a filter is applied to the validation set?

For the word filter, we tested 100 values that divide the validation data into 100 equal-sized percentile groups. For the content filter, integer values from 1 to 100 proved sufficient.

We then applied the elbow method to choose the optimal value—or simply selected the value yielding the best F1 score when the method was not applicable. Figure 30 illustrates this process for the selection of the word filter in the *BERT_all_sources* variant.

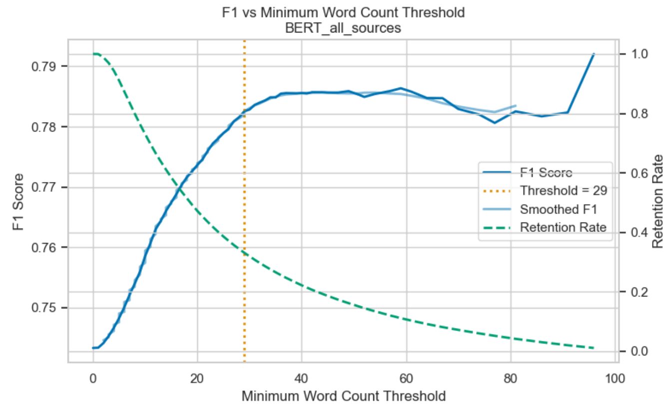

Fig. 30. Word count filter selection for the *BOW_all_sources* variant.

This procedure and the remaining figures are available in the code repository (Appendix A) within the notebook `experiment_result.ipynb`, under the section "Analysis of Incorrectly Predicted Users."

Appendix C
Error Analysis Figures

This procedure is available in the code repository (Appendix A) within the notebook `experiment_result.ipynb`, under the section "Analysis of Incorrectly Predicted Users."

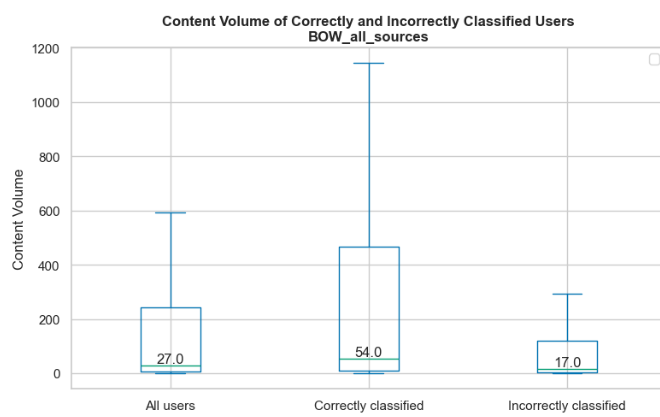

Fig. 31. Content volume (number of posts and comments) of correctly and incorrectly classified users for the *BOW_all_sources* validation set (outliers excluded).

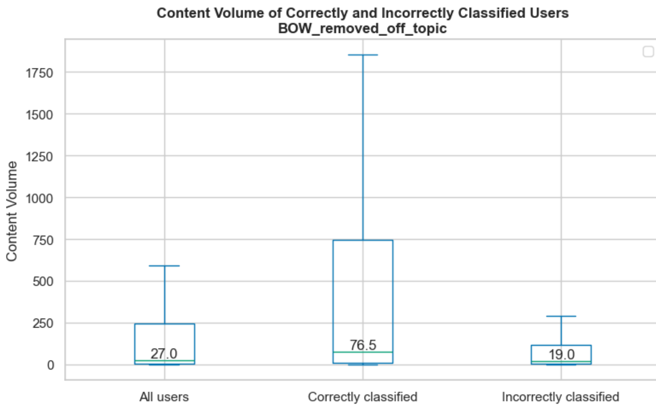

Fig. 32. Content volume (number of posts and comments) of correctly and incorrectly classified users for the *BOW_off_topic* validation set (outliers excluded).

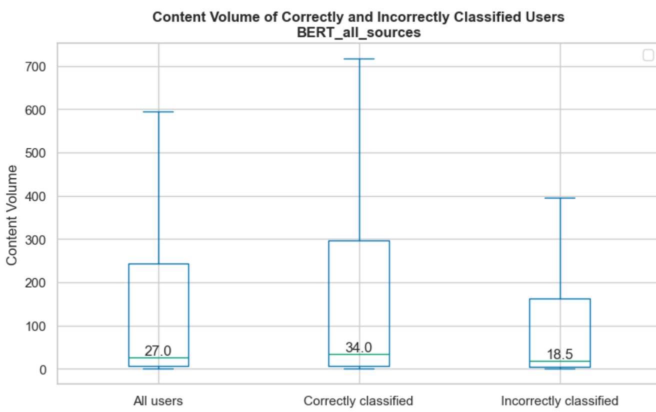

Fig. 33. Content volume (number of posts and comments) of correctly and incorrectly classified users for the *BERT_all_sources* validation set (outliers excluded).

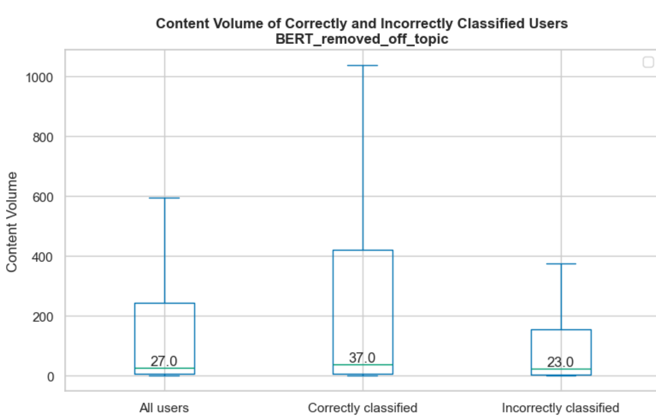

Fig. 34. Content volume (number of posts and comments) of correctly and incorrectly classified users for the *BERT_off_topic* validation set (outliers excluded).